

SAPIENZA UNIVERSITY OF ROME • A.A. 2025/2026

Energy Security Data Warehouse

A Fact Constellation Architecture for Analyzing Geopolitical Risk and Energy Resilience in Europe

Data Management

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The 2022 Energy Shock & Geopolitical Vulnerability

The Geopolitical Shock of 2022

- **Unprecedented Market Volatility:** The outbreak of the Russo-Ukrainian war triggered historic disruptions in natural gas flows, causing TTF gas prices and electricity tariffs to skyrocket across Europe.
- **Asymmetric Exposure:** European member states experienced sharply divergent economic strains based on their baseline reliance on Russian pipeline infrastructure vs LNG alternatives.
- **Decision-Making Gridlock:** Policy-makers faced severely fragmented data: intelligence, market tariffs, and strategic reserves were trapped in isolated institutional silos.
- **Analytical Goal:** Build a unified, time-aware Data Warehouse to quantify the transmission dynamics of international geopolitical risk into real economic energy resilience.



Core Research Question & Native Granularities

CORE RESEARCH QUESTION

“To what extent, and through what temporal dynamics, do international geopolitical tensions co-vary with consumer energy prices, national import dependencies, and strategic oil stockpiles across European nations?”

GEOPOLITICAL RISK

Source: Federal Reserve (Caldara & Iacoviello)

Native Grain: Monthly (1985–2026)

Observations: 499 monthly global rows

Nature: Continuous text-mined news threat index

ENERGY TARIFFS (GAS & POWER)

Source: Eurostat (nrg_pc_202..205)

Native Grain: Half-Yearly (2007–2025)

Observations: 38 semester snapshots per country

Complexity: Household (D1-D3) & Industrial (I1-I6)

DEPENDENCY & OIL STOCKS

Sources: Eurostat nrg_ind_id & nrg_stk_oem

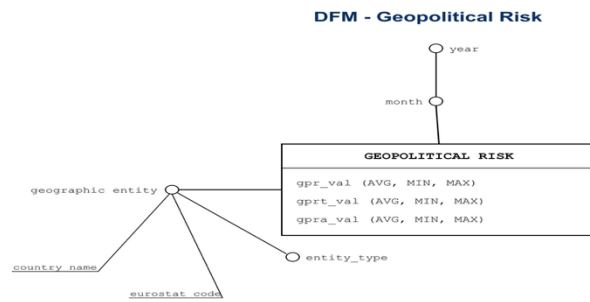
Native Grains: Annual (Dep.) & Monthly (Stocks)

Horizons: 1990–2024 & 2013–2026

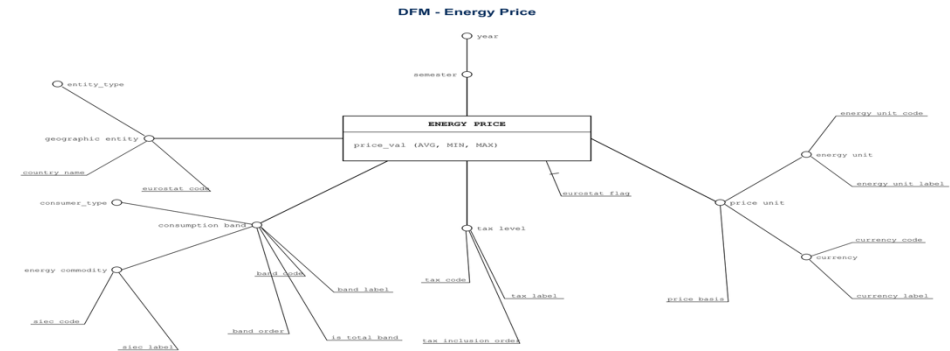
Key Rule: Stock days semi-additive, rates non-additive

The 4 Dimensional Fact Schemes (DFM)

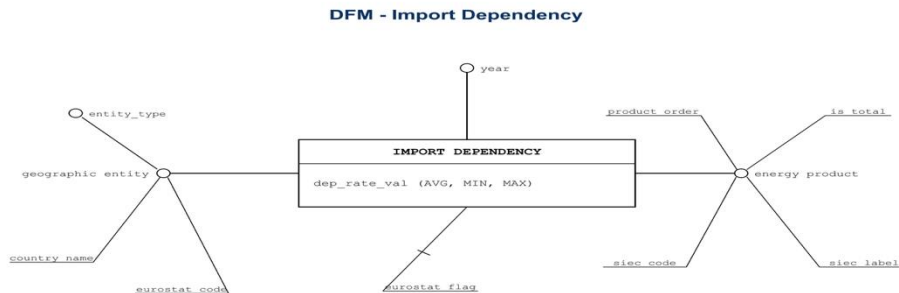
FACT_GPR (Monthly Grain)



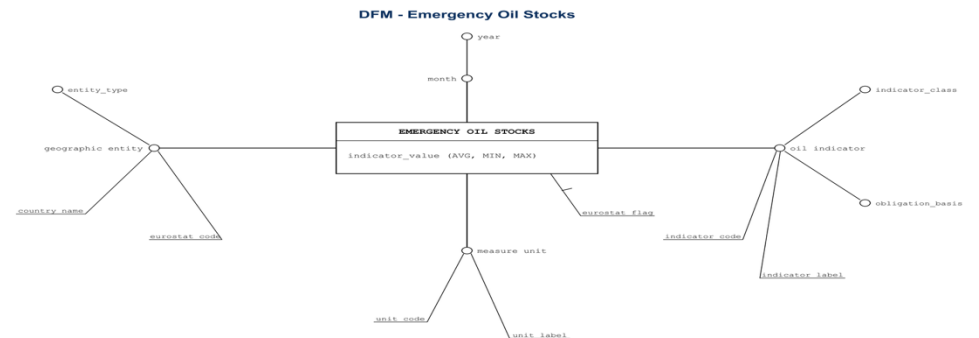
FACT_ENERGY_PRICE (Half-Yearly Grain)



FACT_IMPORT_DEPENDENCY (Annual Grain)



FACT_OIL_STOCKS (Monthly Grain)



Formal Algebra of Measures: Inmon and Dimensional Modeling Principles

1. NON-ADDITIVE MEASURES (UNIT / RATIO QUANTITIES) — SUM IS STRICTLY PROHIBITED

- Applies to: Retail electricity & gas prices (€/kWh), Energy Import Dependency Rate (%), Geopolitical Risk Index (GPR).
- Theoretical Rule: Summing unit prices over semesters or summing percentage rates across nations produces nonsensical values.
- Permitted Operations: Only statistical aggregations (AVG, MIN, MAX) or re-computation as ratio of underlying sums ($\Sigma \text{ revenue} / \Sigma \text{ volume}$).

2. SEMI-ADDITIVE MEASURES (STOCK / LEVEL QUANTITIES) — ADDITIVE ACROSS SPACE, NEVER ACROSS TIME

- Applies to: Emergency Oil Stocks autonomy (days) and physical reserves (kt) in FACT_OIL_STOCKS.
- Theoretical Rule: Reserves can be summed across member states to obtain the total EU buffer, but summing over 12 months is an exam trap.
- Permitted Operations: Over time, only annual averages (AVG), year-end closing levels (LAST), or stress-test minimums (MIN) are mathematically sound.

3. ADDITIVE MEASURES (FLOW QUANTITIES) — UNRESTRICTED SUMMABILITY

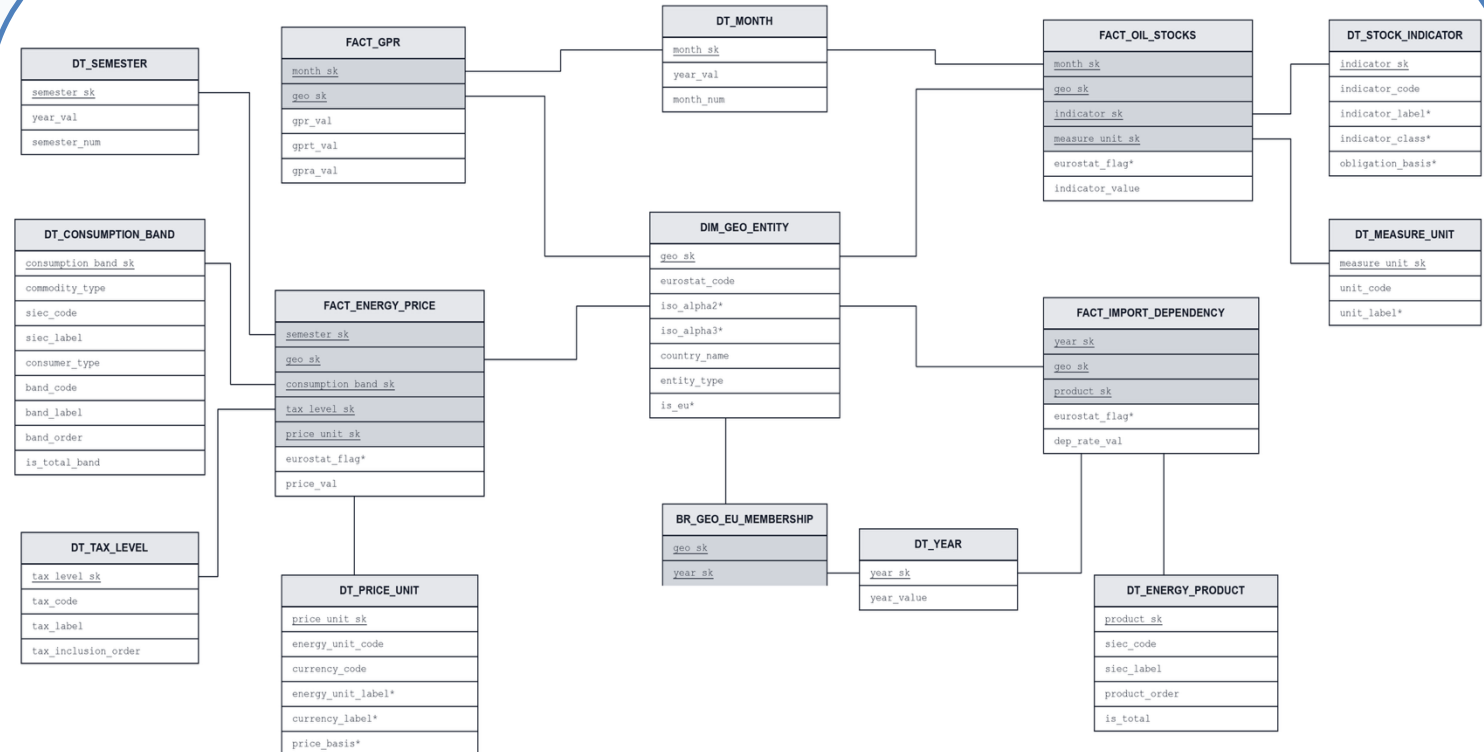
- Applies to: Absolute net imports, physical consumption gigajoules, and total financial revenues.
- Theoretical Rule: Fully additive across all spatial, temporal, and categorical dimensions using standard SUM operators.

Planar Fact Constellation Schema (0 Overlaps & 0 Crossings)

Architecture Rationale

- **Why Fact Constellation?** A single Star Schema would force either artificial replication of annual data or destructive pre-aggregation of monthly risk.
- **Conformed Dimensions:** Unified DIM_GEO_ENTITY (with special surrogate geo_sk=1 for GLOBAL) enables cross-fact drill-across.
- **Surrogate Keys:** Every dimension uses integer surrogate keys (INT/BIGINT) for maximum join efficiency, completely decoupled from source IDs.
- **Planar Layout:** Engineered with 0 line crossings and 0 box overlaps, guaranteeing academic clarity.

STAR SCHEMA - FACT CONSTELLATION



Handling Boundary Evolution: Dynamic EU Membership Bridge

The Static Attribute Trap

- **Changing Borders:** The EU is not a static set of nations. It expanded to 28 members (Croatia in 2013) and shrank with Brexit (UK exit in 2020).
- **SCD Type 1 Flaw:** Storing a simple boolean flag 'is_eu_member' in DIM_GEO_ENTITY would either incorrectly retrodate Brexit to 2010 or falsely include the UK in 2024.
- **Analytical Bias:** Comparing historical EU totals across decades requires tracking the exact legal composition of the union year by year.

The Temporal Bridge: BR_GEO_EU_MEMBERSHIP

- **Bridge Structure:** A dedicated associative table: (geo_sk, year_sk, is_member, accession_year, exit_year).
- **Dual Query Capability:** Analysts can query either the 'EU composition as it was in year T' or the 'constant EU-27 cohort' over time.
- **Example — United Kingdom:** is_member = TRUE for all year_sk <= 2019; is_member = FALSE from 2020 onward.
- **Zero Cartesian Explosion:** Joins directly with fact tables on conformed (geo_sk, year_sk) keys without multiplying measure rows.

Production ETL Pipeline & 12 Automated Integrity Checks

7 MODULAR LOADERS

- **Wide-to-Long:** Eurostat composite headers unpivoted to normalized atomic rows.
- **Metadata Flags:** Flags ('p', 'e', 'b') preserved as descriptive attributes, keeping metrics numeric.
- **Negative Prices:** NL/CZ subsidies audited as legal tax credits, not ingestion bugs.
- **Dependencies:** pandas, psycpg2-binary, xlrd.

IDEMPOTENT UPSERT

- **ON CONFLICT Logic:** Loaders can be interrupted and rerun safely without duplicate rows or key errors.
- **Transaction Safety:** ACID commits and automatic rollbacks on exception.
- **Universal Runners:** One-click automation scripts provided for both macOS/Linux (run_etl.sh) and Windows (run_etl.ps1).
- **Execution Time:** Complete ingestion from empty database executes in ~45 seconds.

12 INTEGRITY AUDITS

After each ETL run, `verify_final_dw.sql` checks key uniqueness, referential integrity, valid value domains, and completeness of the EU-membership bridge. These routine checks help identify incomplete loads or transformation issues before running analytical queries.

Guided 6-Step Didactic OLAP Walkthrough

1. BASE QUERY

Atomic price cell (Country, Semester, Band) -> 1,373 rows

2. ROLL-UP

Ascending hierarchy: Semester -> Year (averaging S1 & S2) -> 701 rows

3. DRILL-DOWN

Introducing consumer type (Household vs Industrial) -> 1,388 rows

4. SLICE

Hyperplane projection fixing Country = 'Italy' (2D plane) -> 36 rows

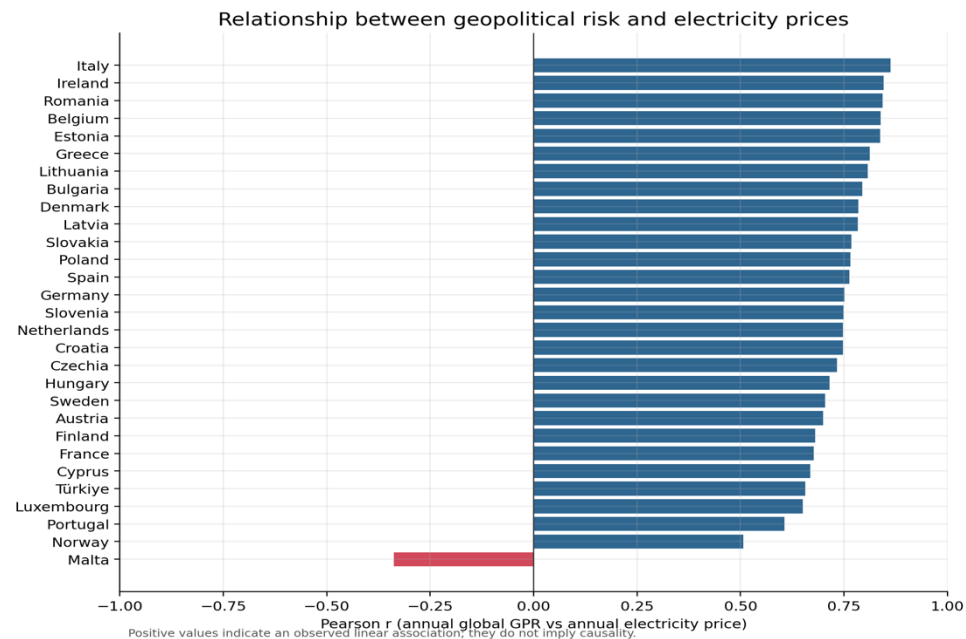
5. DICE

Bounding box: {Italy, Germany} x [2021, 2023] x Households -> 6 rows

6. DRILL-ACROSS

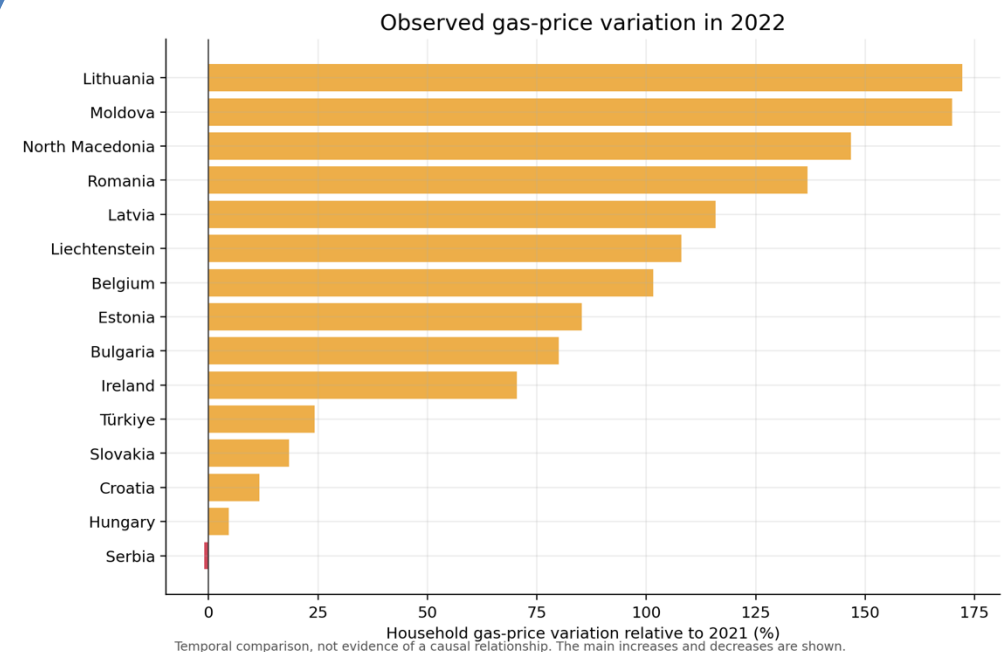
Joining Energy Price + Import Dependency on Conformed Year & Country

Transmission of Geopolitical Tension into Consumer Prices



Query 1: GPR vs Industrial Electricity Correlation

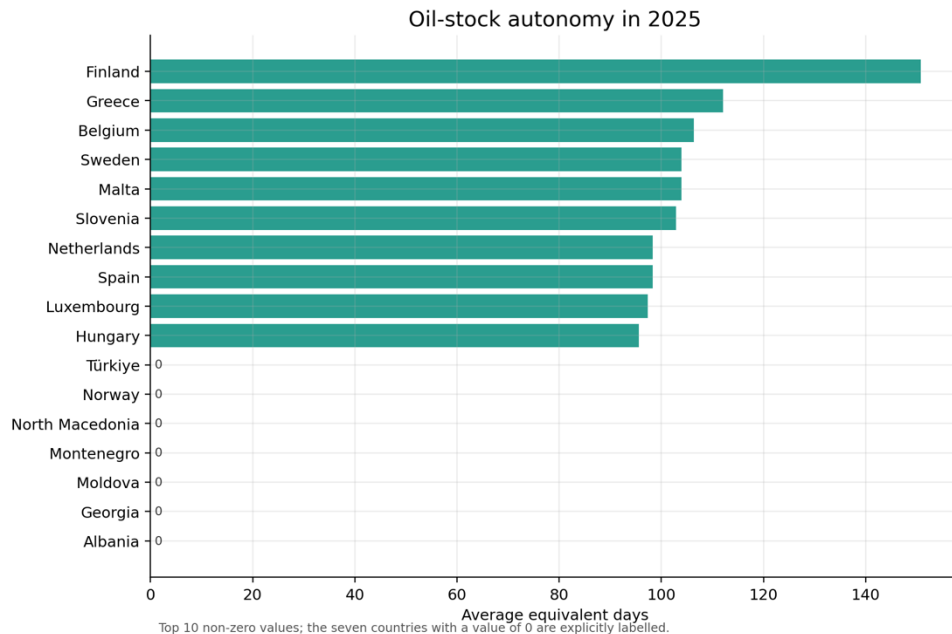
Strong positive correlation across major economies (Italy $r=0.863$, Germany $r=0.751$). Peak tension anticipates retail price peaks by 4–6 months.



Query 5: The 2022 Natural Gas Price Shock

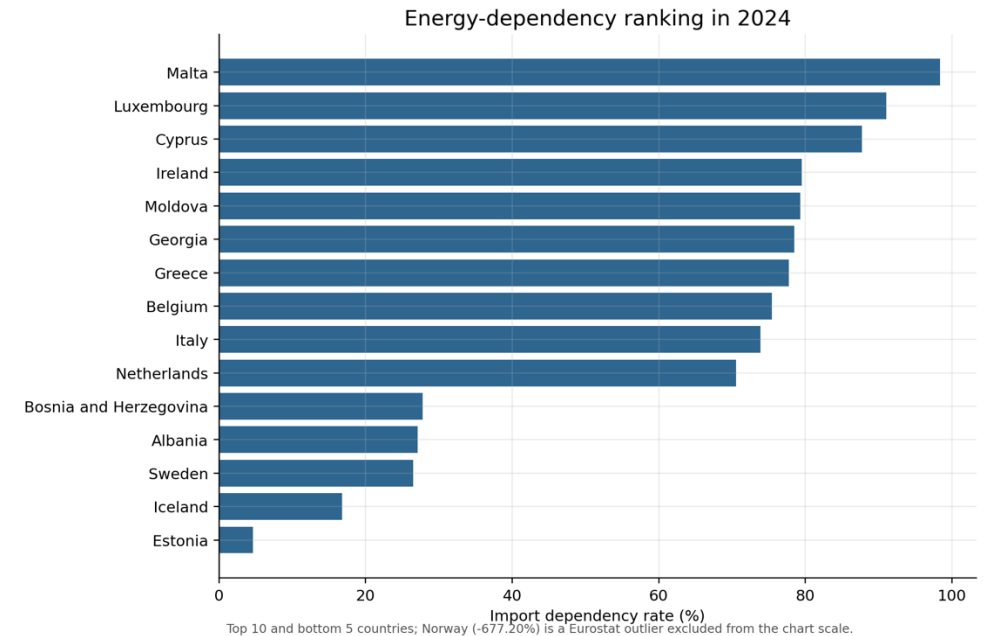
Asymmetric impact of GPR +92% surge: Netherlands (+109%) and Italy (+34%) spiked, while Spain contained increases via Iberian exception and LNG.

Import Dependency Ranking & Oil Stock Autonomy



Query 2: 3-Month Moving Average on Strategic Oil Stocks

Smooths logistic tanker unloads. For Italy in 2022, the average was 86.5 days and MIN() recorded 82.0 days, below the 90-day IEA threshold.



Query 6: 2024 Import Dependency Ranking (DENSE_RANK)

Highlights stark divide: Malta (98.4%) and Luxembourg (91.0%) top vulnerability rankings, contrasted with Norway as net exporter (-677%).

Conclusions, Methodological Limitations & Future Work

KEY ACHIEVEMENTS

- **Rigorous DFM:** Four fact schemes applying the Golfarelli and Rizzi DFM formalisms.
- **Measure Discipline:** Strict adherence to non-additive and semi-additive rules.
- **Fact Constellation:** Zero-crossing planar schema uniting 4 business processes.
- **13 OLAP Queries:** Complete coverage from didactic drill-across to window analytics.

METHODOLOGICAL DEFENSE

- **Correlation != Causality:** Italy shows $r = 0.863$; this describes temporal co-variation, not causation.
- **Market Mechanics:** Merit-order electricity dispatch and take-or-pay gas contracts act as transmission mediators.
- **Granularity Integrity:** Preserving native time frequencies avoided synthetic disaggregation traps.

LIMITS & FUTURE EXTENSIONS

- **Eurostat Latency:** Official annual energy balances are released with a 9–12 month lag.
- **High-Frequency Feeds:** Architecture is ready to integrate daily wholesale power exchange data (GME/EEX).
- **Predictive Views:** Materialized views can feed machine-learning models for early-warning energy price forecasting.